Code Explanation Line by Line (Easy Language)

import numpy as np

Explanation:

- **Imports NumPy library.**
- **Used for:**
  - **Arrays**
  - **Matrix operations**
  - **Mathematical calculations**

Input Patterns (Set A)

```
X = np.array([
    [1, 1, 1],
    [-1, -1, -1],
    [1, -1, 1],
    [-1, 1, -1]
])
```

Explanation:

- **Stores input patterns.**
- **Each row is one input vector.**
- **Values:**
  - **1 = Positive signal**
  - **-1 = Negative signal**

These patterns belong to Set A.

Target Patterns (Set B)

```
Y = np.array([
    [1, 1, 1, 1, 1, 1],
    [-1, -1, -1, -1, -1, -1],
    [1, -1, 1, -1, 1, -1],
    [-1, 1, -1, 1, -1, 1]
])
```

Explanation:

- **Stores associated target patterns.**
- **These patterns belong to Set B.**
- **BAM learns relationship between Set A and Set B.**

Weight Matrix Initialization

W = np.zeros((6, 3))

Explanation:

- **Creates weight matrix of size:**
  - **6 rows**
  - **3 columns**

Initially all values are zero.

Example:

[[0 0 0]
[0 0 0]
...
]

Weight Calculation

for i in range(len(X)):

Explanation:

- **Loops through all training patterns.**

W += np.outer(Y[i], X[i])

Explanation:

- **Calculates outer product between:**
  - **Target pattern**
  - **Input pattern**

Formula:

$$W = \sum Y^T X$$

- **Updates BAM weight matrix.**

This is the learning phase of BAM.

Display Weight Matrix

print("Weight Matrix:")
print(W)

Explanation:

- **Displays final trained weight matrix.**

Example:

[[4 0 4]
[4 0 4]
[0 4 0]
[0 4 0]
[4 0 4]
[4 0 4]]

Activation Function

def activation(x):

Explanation:

- **Defines activation function.**

---

return np.where(x >= 0, 1, -1)

Explanation:

- **If value ≥ 0:**
    - **Output = 1**
- **Else:**
    - **Output = -1**

This is bipolar activation function.

Testing Input Patterns

print("\nTesting for input patterns: Set A")

Explanation:

- **Displays heading for input pattern testing.**

---

for i in range(len(X)):

Explanation:

- **Tests all input patterns one by one.**

Y_test = activation(np.dot(W, X[i]))

Explanation:

- **Multiplies:**
    - **Weight matrix**
    - **Input vector**

Formula:

$$Y = WX$$

- BAM recalls target pattern from input pattern.

---

print(Y_test.reshape(-1,1))

Explanation:

- Displays output as column vector.

Example:

```
[[1]
 [1]
 [1]]
```

Testing Target Patterns

print("\nTesting for target patterns: Set B")

Explanation:

- Displays heading for target pattern testing.

X_test = activation(np.dot(W.T, Y[i]))

Explanation:

- Uses transpose of weight matrix.
- Recalls input pattern from target pattern.

Formula:

$$X = W^T Y$$

This shows bidirectional recall.

---

print(X_test.reshape(-1,1))

Explanation:

- Displays recalled input pattern.

Expected Output

```
Weight Matrix:
[[4 0 4]
 [4 0 4]
 [0 4 0]
 [0 4 0]
 [4 0 4]
 [4 0 4]]
```

Output of Input Pattern 1

```
[[1]
 [1]
 [1]
 [1]
 [1]
 [1]]
```

Output of Input Pattern 2

```
[[-1]
 [-1]
 [-1]
 [-1]
 [-1]
 [-1]]
```

Viva Questions and Answers

1. What is BAM?

BAM stands for Bidirectional Associative Memory.

2. Who developed BAM?

BAM was developed by Bart Kosko.

3. What type of network is BAM?

It is a recurrent neural network.

4. Why is BAM called bidirectional?

Because signals travel in both directions.

5. What is associative memory?

Memory that recalls related patterns.

6. What is hetero-associative memory?

Associating one pattern with another different pattern.

7. What is the role of weight matrix?

Stores association between patterns.

---

8. What is bipolar activation function?

Activation function producing:

- **1**
- **-1**

---

9. What is outer product?

Matrix multiplication between two vectors.

---

10. Why NumPy is used?

For matrix and array operations.

---

11. What is recurrent neural network?

Network having feedback connections.

---

12. What is convergence in BAM?

When output patterns become stable.

---

13. What is Set A?

Input pattern set.

---

14. What is Set B?

Target/output pattern set.

---

15. What does transpose matrix mean?

Rows become columns and columns become rows.

---

16. What is np.dot() used for?

Performs matrix multiplication.

---

17. What is np.outer() used for?

Calculates outer product of vectors.

---

18. Why activation function is required?

To convert net values into binary outputs.

---

19. Can BAM recover noisy data?

Yes.

---

20. What are applications of BAM?

Pattern recognition, memory systems, AI, speech recognition.

Conclusion

Thus, we successfully implemented Bidirectional Associative Memory (BAM) using Python. The BAM network stored associated pattern pairs and recalled output patterns bidirectionally using the weight matrix and activation function.